Compass and Straightedge

The oldest game in math

The one rule. You may only do two things: (1) with the straightedge, draw a line through two points; (2) with the compass, draw a circle from a center and a radius you set on two points. The ruler's numbers are off limits. It is only an edge.

Build 1: the perfect triangle

1. Draw a segment, ends A and B.
2. Compass on A, pencil on B: draw a circle. Same width, compass on B: draw another.
3. Call the top crossing C. Draw AC and BC.

Do it here:

Why are all three sides equal, even though you never measured? Write the reason.

Build 2: cut a segment exactly in half

Compass on A (opened past halfway), arc above and below. Same width on B, arcs crossing. Draw the line through the two crossings. It halves AB *and* makes a right angle.

Build 3: the six-petal flower

Draw a circle. **Keep the compass width fixed.** Step the point around the circle, marking as you go. Connect the marks.

How many steps to get back to start? Count: ________

Why is it *exactly* that number? (Hint: connect the center to two next-door marks. What kind of triangle is that?)

Push further

1. **Bisect an angle** with only compass and straightedge.
2. **Build a square.**
3. **Competition gem.** Someone erased the center of a circle. Find it again with compass and straightedge. (Hint: a chord's perpendicular bisector passes through the center.)
4. **True or false?** You can build each of these with compass and straightedge. Guess first, then ask.

- A regular pentagon (5 sides). T / F
- A regular 7-sided polygon. T / F
- Cutting any angle into three equal parts. T / F